Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750–2024 evidence base for policy analysts and climate negotiators

35+ Bt

global annual fossil fuel and industry CO_2 today

~50%

Asia’s current share of global emissions

150×

gap between high and lowest per capita emitters

Scope: atmospheric milestone, long-run emissions growth, country totals, regional share shifts, per capita inequalities, 2024 annual changes, and energy-mix drivers.

Data sources: Global Carbon Budget 2025; NASA; NOAA Mauna Loa

ATMOSPHERIC MILESTONE

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

NOAA Mauna Loa Observatory measurement, February 2026

~280 ppm

approximate pre-industrial baseline

>50%

increase above the pre-industrial level

1750

pre-industrial baseline
~280 ppm

1958

Mauna Loa record begins;
acceleration becomes visible

2026

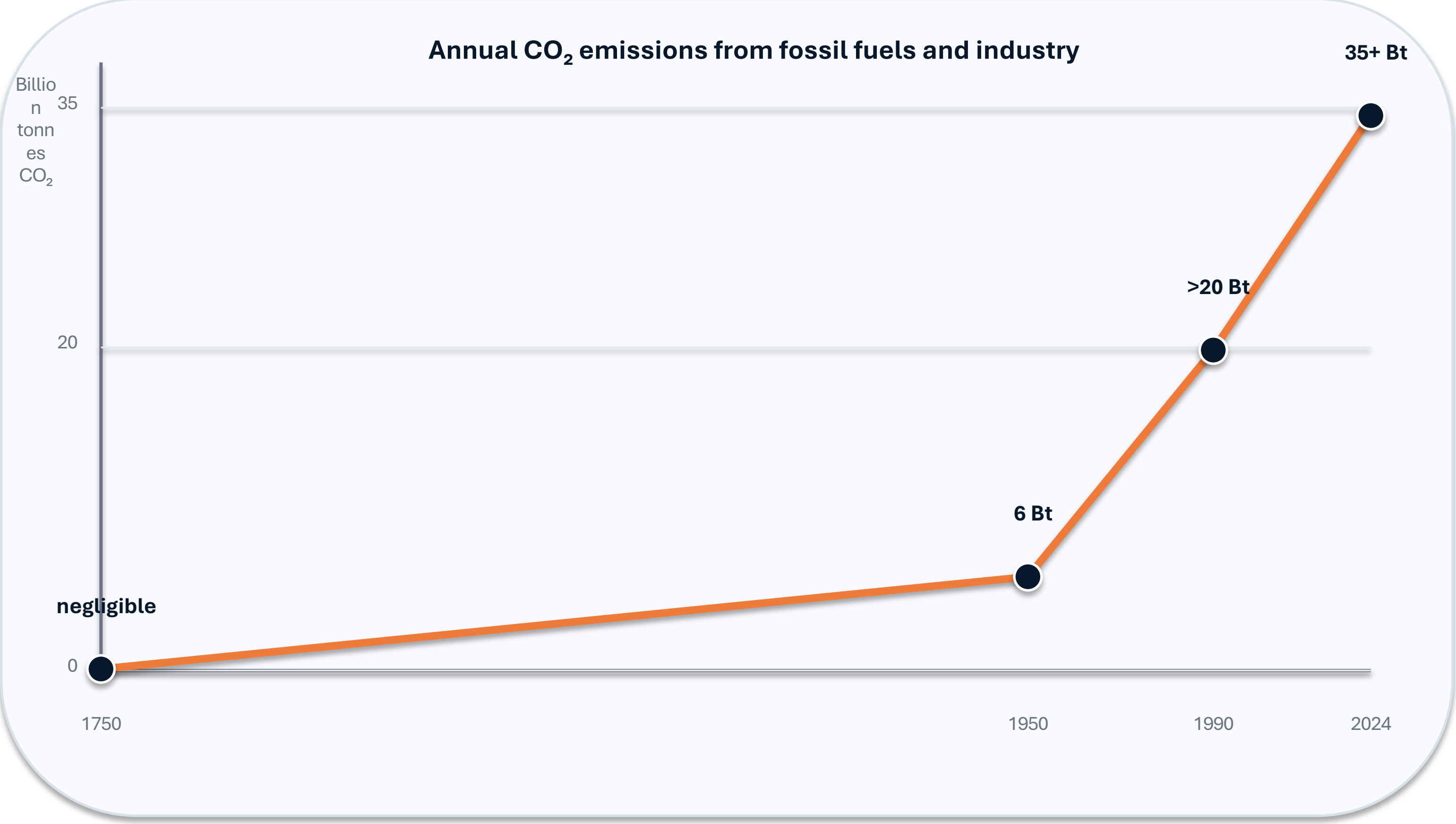
429 ppm in February;
modern concentration milestone

Ground-based and satellite measurements confirm a sharp acceleration linked primarily to burning coal, oil, and natural gas.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

LONG-RUN TRAJECTORY

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



Policy signal

Growth remained relatively slow until the mid-20th century.

Emissions had nearly quadrupled from 1950 to 1990.

Growth has slowed recently, but global emissions have not yet peaked.

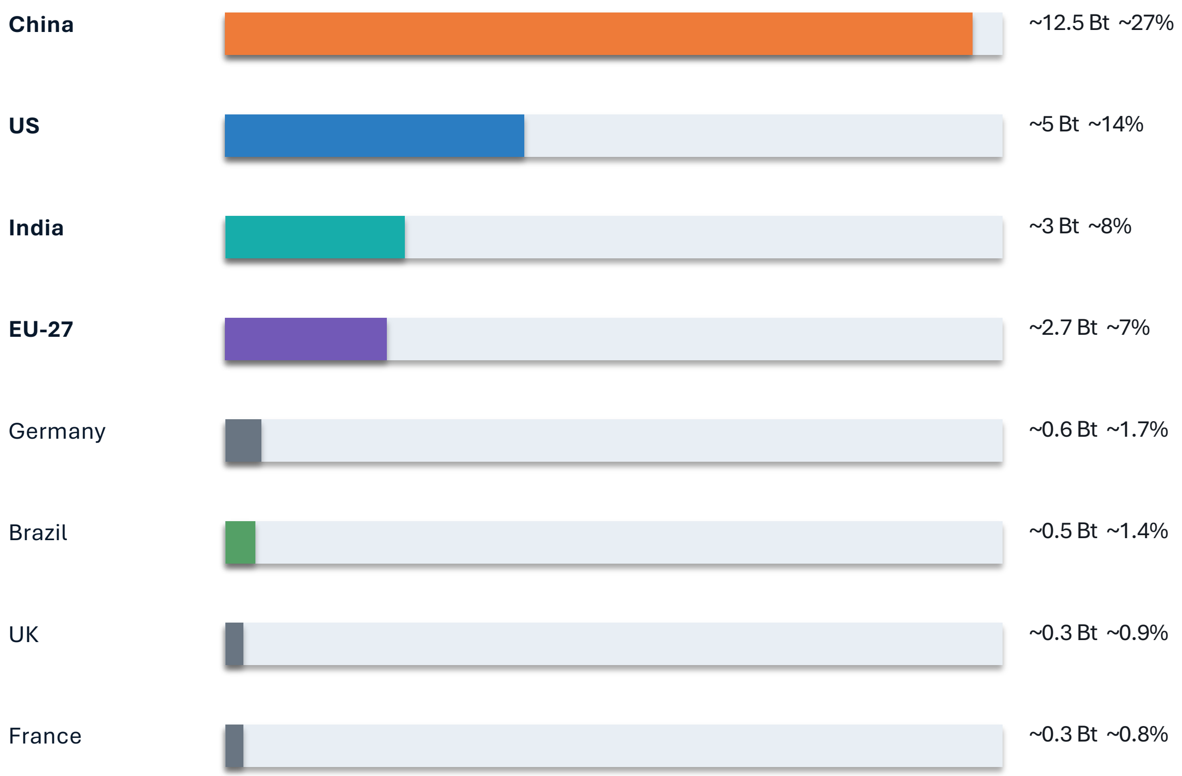
Note: figure_2.jpg was specified in the outline but is not present in the supplied assets; chart uses only source-brief anchor values.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

COUNTRY TOTALS

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Approx. annual CO_2 emissions and global share



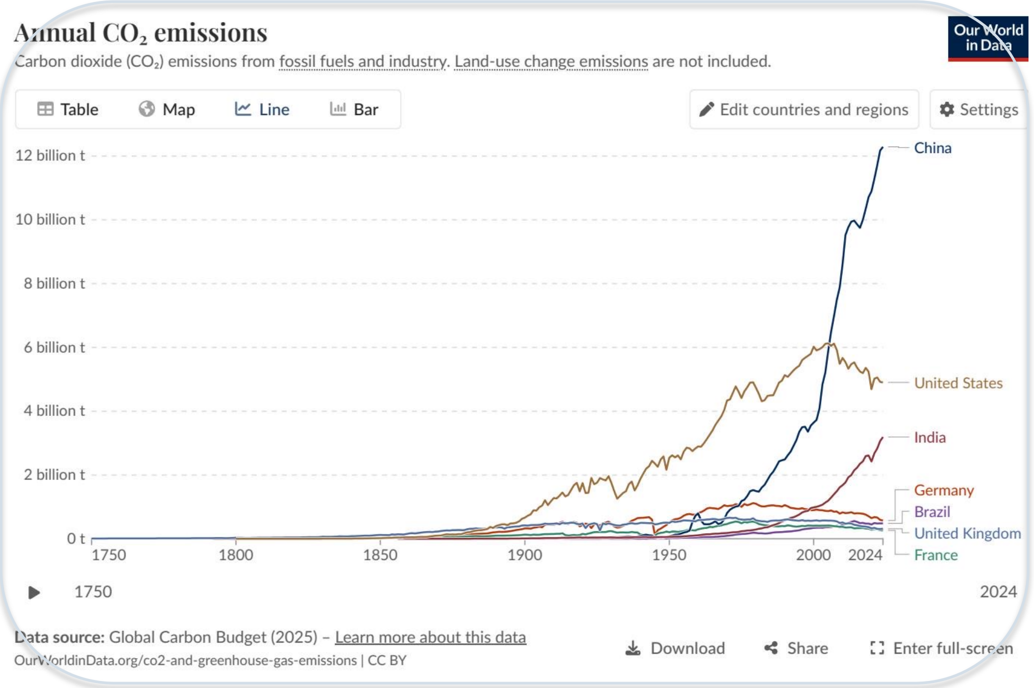


Figure reference: annual CO_2 emissions by country (figure_3.jpg)

China produces over one-quarter of the global total; the United States has declined from its ~6 Bt peak around 2005 to roughly 5 Bt.

India is third-largest and rising fast; Europe's share has fallen substantially.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

REGIONAL SHARES

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

A century-scale relocation of annual emissions

1900

Europe and the US accounted for more than 90% of global emissions.

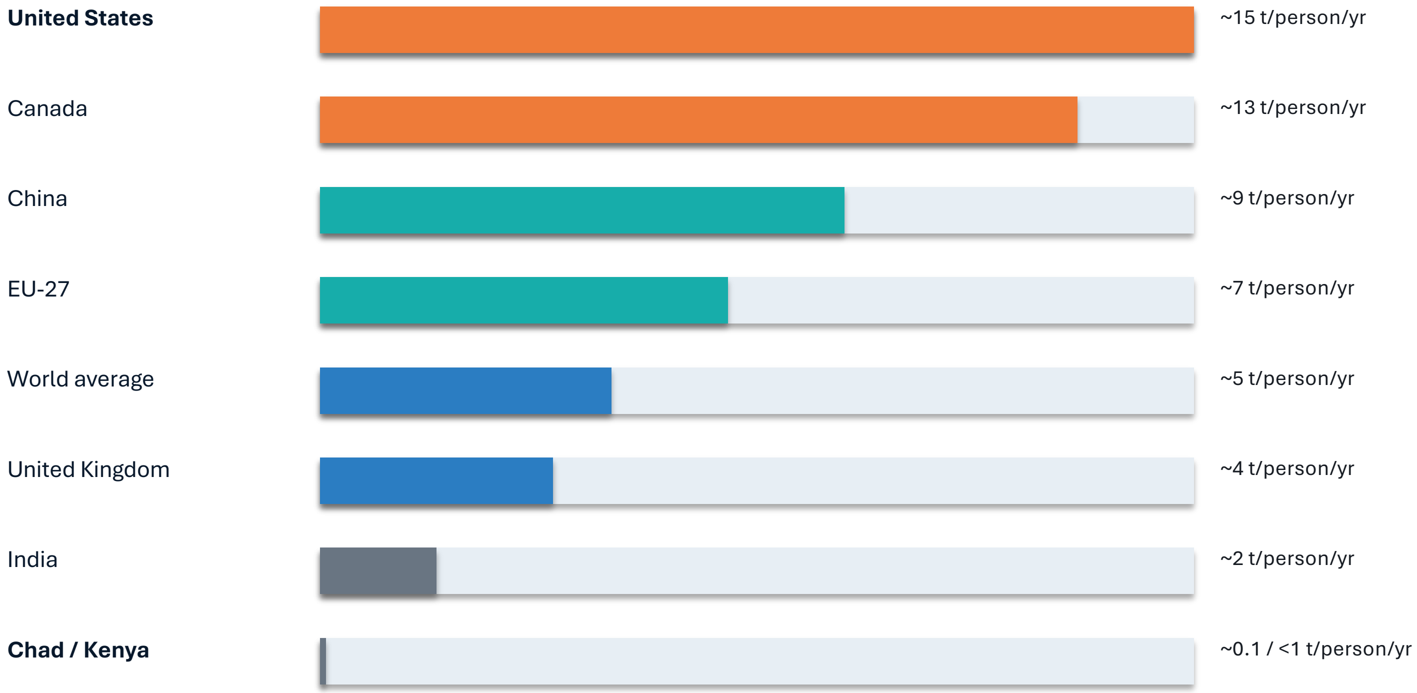
2024

Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.

Negotiation implication: today’s annual emissions center has shifted toward Asia, while historical cumulative responsibility remains concentrated in the US and Europe.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



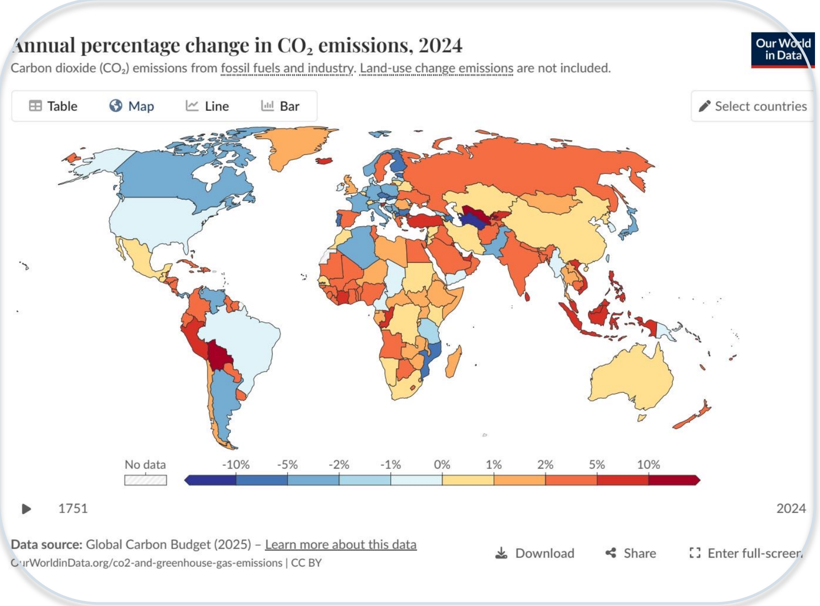


Figure reference: per capita CO_2 emissions over time (figure_1.jpg)

150× gap: a high-emitting person emits in under two days what the lowest emitters release in a year

Among large, populous countries, the US, Australia, and Canada emit roughly three times the global average per person.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

ANNUAL PERCENTAGE CHANGE

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

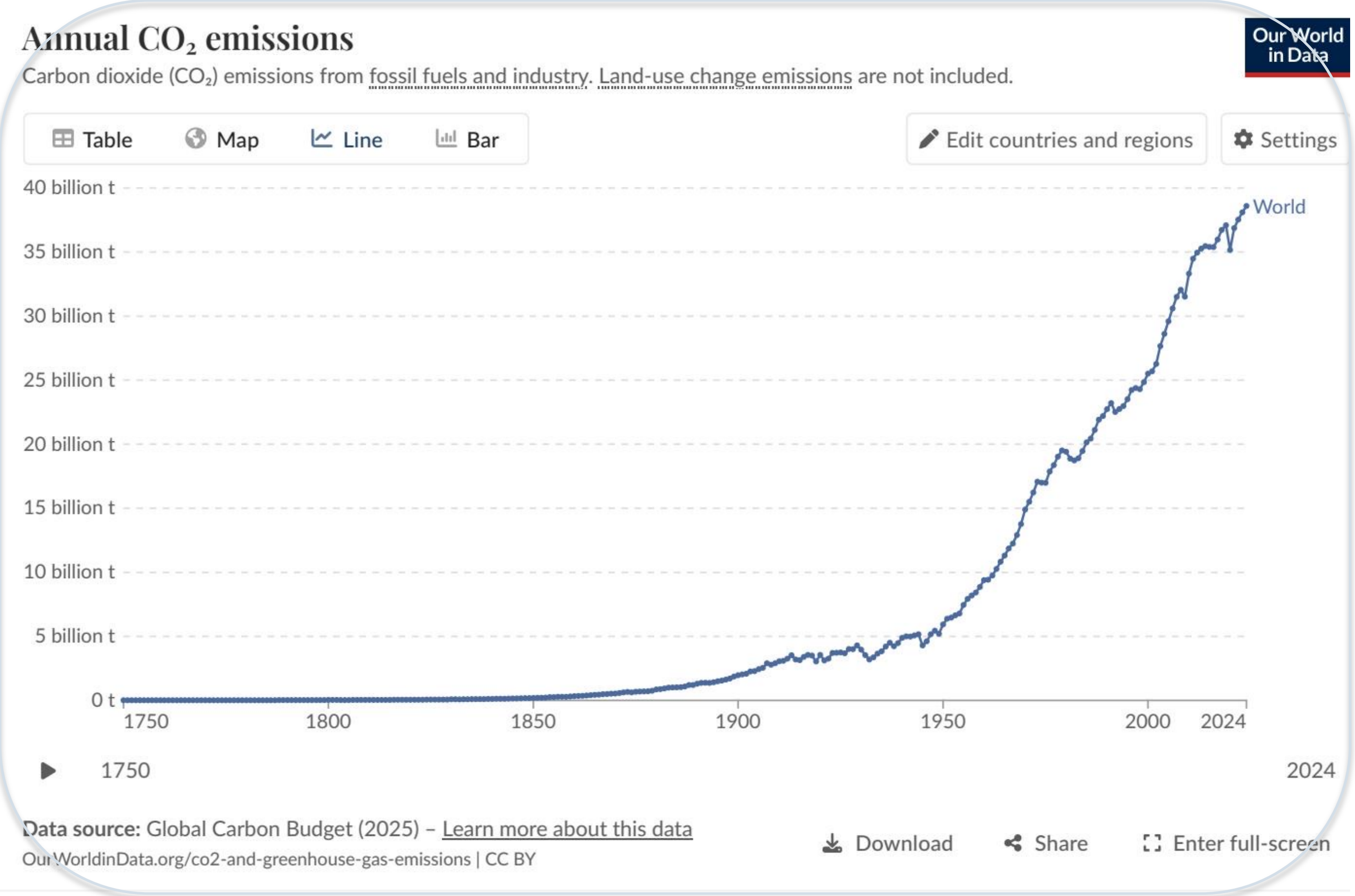


Figure reference: annual percentage change in CO_2 emissions, 2024 (figure_4.jpg)

Map reading guide

−1% to −5% — US and much of Europe show declines

+2% to +10%+ — Russia, parts of Southeast Asia, and several African nations increase

>+10% — Some South American countries show very large percentage increases

Mixed trajectory

Declines in mature economies coexist with rapid growth across parts of Eurasia, Africa, Southeast Asia, and selected South American countries.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions because electricity and industrial energy sources differ.

Country group	Energy-source profile in source	Per capita CO_2 in source	Policy interpretation
France	Higher nuclear and renewable shares	~4 t/person/year	Lower per capita emissions despite high living standards
United Kingdom	Higher nuclear and renewable shares	~4 t/person/year	Lower than other similarly prosperous fossil-heavy systems
Germany	Higher fossil fuel share	Higher than France/UK (exact value not specified)	Energy mix raises per capita outcomes
Netherlands	Higher fossil fuel share	Higher than France/UK (exact value not specified)	Energy mix raises per capita outcomes

Prosperity alone + **Energy mix** → **Different per capita CO_2 outcomes**

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)

SYNTHESIS

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global emissions exceed 35 Bt/year and have not yet peaked.
2. China emits over 25% of the global total — more than double the US.
3. Regional shares shifted from >90% Europe + US in 1900 to Asia at ~50% today.
4. Per capita inequalities remain extreme: a 150× gap separates high and lowest emitters.
5. Energy mix and policy choices shape per capita outcomes, not prosperity alone.

Negotiation frame: annual emissions now center on emerging Asian growth, while historic high-income responsibility and per-capita inequity remain central to burden-sharing debates.

Source: NASA; Our World in Data — CO_2 and Greenhouse Gas Emissions (Global Carbon Budget 2025)